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Ionogel based on biopolymer-silica interpenetrated networks: dynamics of confined ionic liquid with lithium salt

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Obtaining solid-state electrolytes with good electrochemical performances remains challenging. Ionogels, i.e. solid host networks confining an ionic liquid, are promising as they keep the macroscopic properties of the liquid. However, confinement of an ionic liquid can imply important changes in its molecular dynamics, depending on the route of synthesis and on the confining network. We studied this effect on an imidazolium based ionic liquid with its lithium salt confined in a hybrid biopolymer-silica matrix. Dynamics of bulk and confined solution was probed by quasi-elastic neutron scattering (QENS) which revealed a weakly slowed dynamics of imidazolium-based ionic liquid inside the polymer-silica host network.

Introduction

Ionic liquids (IL) have noteworthy and well-known properties such as very low volatility, good chemical and thermal stability, wide electrochemical window and good ionic conductivity. These properties make them particularly promising for developing safer electrochemical energy storage devices (double layer capacitors, lithium batteries and solar cells,) [1].

One way to take advantage of these properties is to confine ILs within host networks, like polymers, mesoporous oxides or hybrid organic-inorganic matrices. Such solid-state materials can be obtained either by soaking a previously prepared host network in an IL [2,3] or more efficiently by a one-step sol-gel process leading to ionogels [4]. Silica-based ionogels, with or without added polymer, are solid materials with liquid-like properties [5] whose efficiency as solid electrolyte have been demonstrated in supercapacitors [6] and lithium batteries [7].

Hybrid polymer-silica ionogels are especially promising as solid electrolyte, because they offer more flexible materials [8] than brittle silica-based ionogels [9–11]. Flexibility as well as good resistance to lengthening would make them easier to implement in industrial production of devices for energy storage. More specifically related to the present study, a previous work [12] has reported ionogels which host network is made of cellulosic derivative, i.e. a silanised hydroxypropyl methylcellulose (Si-HPMC), covalently linked to silica nanofibers (NFs) via siloxane bridges. The corresponding ionogel is obtained after several exchanges of solvents, leading to a hydrophobic IL confined in a hydrophilic host network. With an amount of NFs above its percolation threshold, such solid electrolyte has shown an enhanced ionic conductivity compared to that of the bulk IL[12].

It is noteworthy that phase transitions of ILs are usually modified when they are confined in silica matrices, resulting in liquid-like behaviour below the solidification temperature of the neat IL [5,12,13]. The dynamics of molecular liquids is prone to be modified under confinement. In the case of ILs, literature reports either fastened [14,15] or lowered [16] molecular dynamics inside different kinds of host network.

To the best of our knowledge, no study reports on the effect of confinement on the local dynamics of the IL in presence of lithium salt inside a hybrid host network. The aim of this work was to study the influence of confinement on the dynamics of a model IL, i.e. 1-butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide (BMImTFSI) with 0.5 M LiTFSI salt, namely BMIm Li TFSI. The previously described hybrid biopolymer – silica (Si-HPMC/NFs) based host network was chosen as confining matrix, because of its interesting higher ionic conductivity than that of the bulk IL. However, the previous process to obtain this ionogel was time-consuming [12]. We thus first describe here a simpler route to obtain this ionogel and to allow its shaping in thin films. Then the dynamics of the IL with its lithium salt confined within a hybrid biopolymer-silica matrix has been probed by means of quasi-elastic neutron scattering (QENS) and the results have been compared to its bulk counterpart.

Experimental

Ionic liquid and ternary solution

1-Butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide BMImTFSI was synthesized as a colourless transparent liquid, and purified, according to the literature [17]. 1-methylimidazole

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was reacted with a slight excess of 1-bromobutane in a roundbottom flask and left refluxing for 2 h to produce 1-butyl-3-methylimidazolium bromide. The corresponding TFSI salt was then prepared by metathesis of BMIm Br in water with lithium bis(trifluoromethylsulfonyl)imide (Li TFSI). The final IL was checked by 1H-NMR. The ternary solution was obtained by adding Li TFSI (0.758 g) to BMim TFSI (6.822 g) resulting in a solution containing 0.5 M of lithium salt namely BMIm Li TFSI.

Ionogel Synthesis

The hybrid ionogel was synthesized by a sol-gel process involving the mixing of two solutions, inspired from a previous work [18]. The solution A was obtained by dissolving Si-HPMC polymer in 0.2 M NaOH aqueous solution (30.9 mg mL^{-1}, pH 12.5); then two successive dialyses with molecular weight cut off at 6–8 kDa were performed in 0.09 M NaOH aqueous solution. These dialyses allow eliminating the non-grafted 3-glycidoxypropyltrimethoxysilane, used for silane grafting onto the HPMC [18]. In solution B, 60 mg silica NFs was dispersed by ultrasounds in 2.36 mL of a mixture of acetonitrile and formic acid. 1 g of BMIm Li TFSI was then added. The ionogel sol precursor was then obtained by mixing 1.4 g of solution A with solution B. Gelation of the sol occurred in four hours.

Materials characterisation

Thermogravimetric analyses (TG-DSC 111, Setaram) were carried out at a heating rate of 5 $K.min^{-1}$ from room temperature to 1073 K, under air flow. Phase transitions were probed by differential scanning calorimetry (DSC) (Q20, TA instrument) and the data were analysed with TA Universal Analysis software. Samples of 5-8 mg were placed in hermetically sealed aluminium pans. They were dried at 333 K under vacuum during at least 12 h before measurements. Pans were exposed to a helium flow atmosphere. Samples were cooled down (quenched) to 120 K, followed by heating from 120 to 373 K at a rate of 10 $K.min^{-1}$. Temperatures of glass-transition, crystallisation and melting were determined on heating at the onset of each transition.

QENS experiments

QENS experiments have been performed on the cold neutron time-of-flight spectrometer FOCUS [19] at the Swiss spallation neutron source SINQ, Paul Scherrer Institute, Villigen, Switzerland. For these experiments, we used neutrons with an incident wavelength of λ = 5.76 Å. The energy resolution (FWHM) at this wavelength is $\Delta E \approx 45$ µeV. Spectra were recorded for 4 different temperatures T = 30, 233, 300 and 333 K. Data obtained at 30 K were used as the resolution function. Samples were dried at 333 K under vacuum during at least 12 h before measurements. We used flat sample containers made of aluminium, which suits with 200 µm thick samples. This thickness guarantees neutron transmission of >90% and thus suppresses multiple scattering. The measured intensities of a vanadium standard with the same shape as the sample have been used to calibrate the detectors. We employed measurement times of about 6 h for each temperature. The DAVE software package [20] was used for data reduction and analysis. Data were binned into 14 groups of constant Q values; the energy range was set to ±1 meV.

Results and Discussion

Biopolymer-silica based ionogel

The use of modified HPMC with pendant silanol moieties together with NFs allows forming a hybrid network cross-linked via Si-O-Si bridges. These silica NFs are around 50 nm in diameter and 500 nm in length (synthesis described elsewhere [21]). Such a form factor allows reaching a percolating silica network with a small amount of NFs (below 3 wt% de the whole material), resulting in a remarkable change in macroscopic properties, such as a sharp increase of both Young modulus and ionic conductivity [12,18]. Until now, this material was first synthesised in aqueous medium and ionogel was then obtained by liquids exchange. This process was time-consuming and not suitable to shape thin films.

We have developed a process which allows the cross-linking of the Si-HPMC/NFs network directly in a hydrophobic ionic liquid medium: NFs were dispersed in acetonitrile, formic acid and IL. Acetonitrile was used to make possible the mixing of aqueous Si-HPMC solution with hydrophobic IL, while formic acid would trigger the polycondensation via Si-O-Si bridges [9,22]. The Si-HPMC solution was then added to the NFs suspension leading to a gelation of the sol within four hours. The final composition of this material was checked by thermogravimetric analysis. This route allowed reaching an IL-Li content as high as 91 wt%, within a confining network made of 57 and 43 wt% of Si-HPMC and NFs respectively: the composition could thus be either summarized for [Si-HPMC / NFs] / IL-Li as [57/43] / 91 wt% or for Si-HPMC / NFs / IL-Li as 5.1 / 3.9 / 91 wt%.

This ionogel has been characterized by DSC and its phase transitions have been compared to its bulk counterpart (Fig.1).

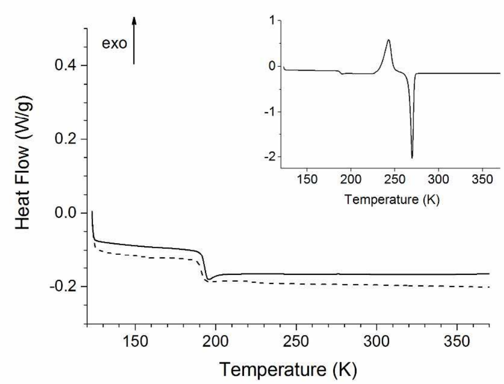

Fig. 1 DSC thermograms of bulk BMIm Li TFSI (solid line) and the corresponding ionogel Si-HPMC / NFs (dots). Insert: pristine ionic liquid BMIm TFSI. Data were recorded at 10 $K.min^{-1}$ after quenching at 123 K.

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Table 1 Thermal characteristics extracted from DSC measurements.

Sample	T_g (K)	ΔCp (J/g.K)	T_f (K)
BMIm TFSI	186.2	0.37	266.4
BMIm Li TFSI	191.0	0.39	N.A.
Si-HPMC-NFs ionogel	190.5	0.34[a]	N.A.

[a] This value is reported to the mass of IL. N.A. stands for not applicable.

Both samples present a glass transition around 190 K (Table 1), even if this transition seems to be very slightly shifted to lower temperatures in the case of the Si-HPMC/NFs ionogel. This is a clear indication that the confinement inside the solid ionogel retains the thermal properties of the IL-Li.

Nevertheless, it is worth to point out that confinement reduces the amplitude of *ΔCp* at the glass transition associated with reduced entropy of the confined IL.

It has also to be noted that thermal measurements revealed the strong influence of Li cation on the physical properties of the IL. The solid/liquid transition, which occurred at 266 K for pristine IL, has disappeared with the incorporation of 0.5 M Li TFSI, while its glass transition has been shifted by 5 K towards higher temperatures (Fig. 1, insert).

Influence of lithium ions on the dynamics of $BMim^+$ cations

Quasi-elastic neutron scattering (QENS) is an appropriate technique to study dynamical processes in condensed matter on the length scales below one nm and time scales of a few ps. It allows investigating diffusion, relaxation and/or jump diffusive motions and extracting corresponding quantitative parameters (diffusion coefficients, jump lengths and characteristic times). QENS is very sensitive to isotopes with large incoherent neutron scattering cross section, typically samples containing many hydrogen atoms. As this technique cannot give direct information on the dynamics of $TFSI^-$ and Li^+ ions (they do not contain any proton), we had only probed the dynamics of imidazolium cations. In general, the incoherent dynamic structure factor can be expressed as:

$$\mathbf{S(Q,\omega) = e^{-\frac{\langle u^2\rangle Q^2}{3}}\{A_0(Q)\delta(\omega) + [1 - A_0(Q)]L(\omega,\Gamma)\}} \quad (1)$$

Where $A_0(Q)$ is the elastic incoherent structure factor, δ is the Kronecker delta function and $L(\omega,\Gamma)$ is a Lorentzian function that describes the quasi-elastic spectral component, with full width at half maximum (FWHM) given by Γ. The mean square displacement $\langle u^2\rangle$ is a measure of the spatial extent of the motional process responsible of the quasi-elastic broadening. The analysis of the elastic intensity at these four temperatures has allowed determining the mean-square displacement (Fig. 2-b) according to the following equation 2:

$$\mathbf{I_{elastic} = e^{-\frac{\langle u^2\rangle Q^2}{3}}}$$

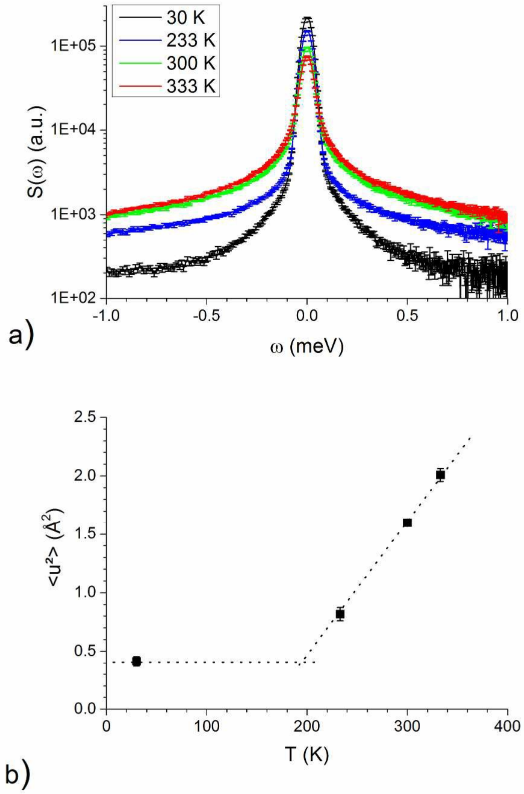

Fig. 2 a) QENS spectra of BMIm Li TFSI obtained at T = 30, 233, 300 and 333 K. Intensity has been summed over the whole Q range. b) Mean-square displacement of BMIm Li TFSI as a function of temperature. Dotted lines are guide for the eye.

Usually, no molecular motions occur at temperatures below 110 K and thus the mean-square displacement does not vary in this temperature range [23]. The lower part of the dotted line is just a guide for the eye: despite the uncertainty, a slope break seems to occur below 233 K, highlighting the transition from pure vibrational motions to translational diffusion upon heating. Nevertheless, it is worth pointing out that the observed behaviour is coherent with DSC measurements made on BMIm Li TFSI, which showed the glass transition at 191 K (Fig.1).

However, there should be no pure elastic scattering from bulk liquids and experimental data are usually fitted with a combination of two Lorentzian functions, including the case of bulk IL [15,16,24–26]. The total incoherent structure factor $S(Q,\omega)$ can therefore be written:

$$\boldsymbol{S(Q,\omega) = e^{-\frac{\langle u^2\rangle Q^2}{3}}\{A_0(Q)L(\omega,\Gamma_1) + [1 - A_0(Q)]L(\omega,\Gamma_2+\Gamma_1)\}} \quad (3)$$

QENS experiments on bulk and ionogel-confined BMIm Li TFSI were realized at temperatures of 30, 233, 300 and 333 K. From 233 K to 333 K, the ternary IL BMIm Li TFSI is in a liquid

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phase. Typical spectra *S(Q, ω)* of the bulk solution is presented for the four investigated temperatures (Fig. 2-a). The base of the peak, which corresponds to the quasi-elastic part of the spectra, broadens when the temperature increases. This corresponds to an increase of the molecular dynamics: at 30 K the signal is considered completely elastic and due to local vibrational motions, while at higher temperatures quasi-elastic broadening corresponds to translational diffusion.

In one hand, the broadest Lorentzian (around 1 meV) usually characterizes the fast and localised dynamics connected to such motions as conformational changes or molecule libration In the other hand, the narrowest Lorentzian function, via its FWHM (Γ_1), characterizes the translational diffusion of the cation (Fig. 2-a). It ranges around 0.1 meV for these experiments. At 300 and 333 K, the scattering function of bulk BMIm Li TFSI has been fitted satisfactorily with this model of two Lorentzian functions. However, it did not match for the scattering function measured at 233K. The dynamics of BMIm Li TFSI should be too slow at this temperature with regard to the energy resolution of 45 µeV at λ = 5.76 Å.

The FWHM of the broadest Lorentzian was Q-independent for these three temperatures (results not shown): this is characteristic of localised non-diffusive dynamics, and not in the scope of this paper.

In the case of simple translational diffusion (the narrowest Lorentzian function), the FWHM of the translational term (Γ_1) has a quadratic Q-dependence. At large Q values, dynamics can be often described by the rapid jump diffusive model (see equation 4) [27]:

$$\boldsymbol{\Gamma = \frac{2\hbar DQ^2}{1 + DQ^2\tau_0}} \tag{4}$$

Where τ_0 is the characteristic residence time between jumps of a diffusive particle with a diffusion coefficient $D = \langle l^2\rangle/6\,\tau_0$, where $\langle l^2\rangle$ is the mean square jump length [15]. This last model allowed fitting data with an acceptable precision. Values of fitted parameters are reported in Table 2. The diffusion coefficient of $BMIm^+$ in BMIm Li TFSI has the same magnitude as those of others TFSI-based ILs in their liquid phase [24,28,29]. It tends to be slightly lower than neat BMim TFSI along with a decrease of the characteristic time for $BMIm^+$, indicative of more frequent jumps. Despite this, $BMim^+$ cations appears to be less moving in presence of lithium salt regarding the values of mean jump length (Table 2).

In our case, the presence of lithium salt slightly lowers the dynamics of the $BMIm^+$ cation of pristine IL. This is coherent with others experimental facts such as the increase of both density and viscosity [30] and upheaval of thermal transitions like the disappearance of crystallisation and melting (Fig. 1, insert). Such phenomena have been previously reported with various kinds of IL: imidazolium [31] or pyrrolidinium cations [32] and more recently with phosphonium bis(fluorosulfonyl)imide (FSI) [33].

Table 2 Values of the diffusion coefficient *D* and the characteristic time τ_0 of bulk and confined BMIm Li TFSI at 300 K and 333 K.

Sample	T (K)	*D* (10^{-10} $m^2.s^{-1}$)	τ_0 (ps)	<l> (Å)
Bulk BMim TFSI [a]	298	2.4 (± 0.1)	26 (±3)	1.9
Bulk BMim TFSI [b]	290	3.06 (±0.24)	30	2.3
Bulk BMIm Li TFSI	300	1.30 (±0.07)	10 (±3)	0.9
	333	2.62 (±0.06)	7 (±1)	1.0
Confined BMIm Li TFSI	300	1.41 (±0.05)	33 (±3)	1.7
	333	2.33 (±0.06)	16 (±2)	1.5

[a] from reference 24; [b] from reference 11.

MD simulations has helped to investigate the liquid structure of IL in presence of lithium ions [30,34]. Monteiro et al [30] reported the formation of a stiffer structure composed of 'strong' (Li^+ with $TFSI^-$) and 'weak' (imidazolium-TFSI) aggregates. In the meantime, several studies [34,35] have proposed that the ion Li^+ together with $TFSI^-$ anions forms $[Li(TFSI)_2]^-$ clusters – and even $[Li(TFSI)_4]^{3-}$ clusters for solutions with high content of LiTFSI [35] either for pyrrolidinium or imidazolium derivatives. In our case the weakening of BMim-TFSI interactions by Li-TFSI ones quench the long-range ordering, *i.e.* the crystallisation/melting of BMim TFSI. The bulk solution has thus a greater ability of glass forming and the molecular dynamics of the IL is slowed down. Such disturbance caused by lithium ions must influence the value of the coefficient diffusion *D*, the characteristic residence time τ_0, and thus the mean square jump length $\langle l^2\rangle$. PFG-NMR studies on phosphonium FSI with corresponding lithium salt also evidenced a significant decreasing mobility of cation and anion, lithium ions becoming much more mobile at relevant concentrations in the IL [33].

Dynamics of confined $BMim^+$ cations

When the IL is confined in the polymer-silica host network, the QENS peak becomes slightly sharper than these of bulk IL at the same temperature (Fig. 3).

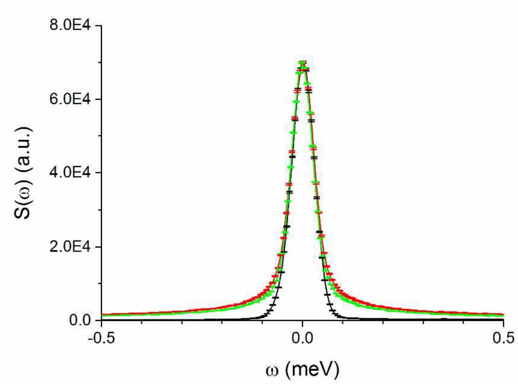

Fig.3 QENS spectra of BMIm Li TFSI (red) and ionogel (green) obtained at T = 333 K normalised to the vanadium spectrum (black). Intensity has been summed over the whole Q range.

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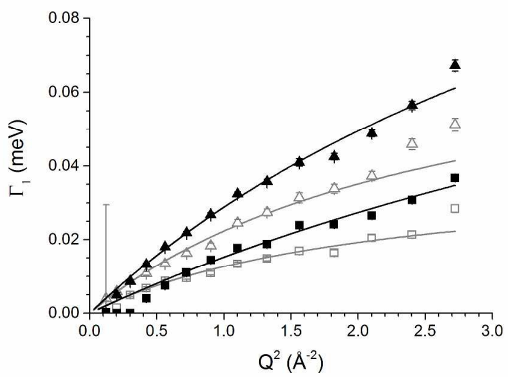

Fig. 4 FWHM of the diffusive component (Γ_l) versus Q^2 for bulk (black symbols) and ionogel-confined (empty symbols) BMIm Li TFSI at 300 K (■) and 333 K (▲). Solid lines are fitting curves of experimental data.

It suggests that the mean dynamics of $BMim^+$ cations is slightly slowed down inside the confining network. Two Lorentzian functions also fitted well to the experimental data recorded at 300 and 333 K. The narrow Lorentzian component corresponds to diffusive motion [24,28]. The FWHMs extracted from the fits of this narrow component Γ_1 have been represented as a function of Q^2 (Fig. 4). For the same temperature, the FWHM of confined IL is always smaller than that of the bulk one, indicating a slight slowdown of the dynamics of the $BMIm^+$ cation inside the Si-HPMC/NFs host network. This effect is more distinct at 333 K. The fit of experimental data shows the decrease of D and the increase of τ_0 when the IL is confined (Table 2) which is consistent with the first qualitative observations (Fig. 4). While confinement can induce drastic changes in physical properties of confined molecules [36,37], the slowdown of the confined imidazolium cation herein remains limited within this polymer-silica host network.

The dynamics of ILs appears however to be more complex than simple molecular liquids usually described by a two-Lorentzian functions model. Selective deuteration of either the alkyl chain or the ring of the cation has allowed disentangling and highlighting different kinds of motions, such as conformational relaxations or translational motions [28,29,38]. In particular, conformational motions of alkyl functional group, like rotational reorientation or dihedral relaxation, can interfere with the signal stemming from the global dynamics of $BMIm^+$ cation [28]. This is especially true when the length of the alkyl side-chain increases: for example the deuteration of octyl side chain of imidazolium bromide reduces the measured value of D by almost two-third compared to the fully protonated counterpart [28]. Garaga et al [39] have also observed this phenomenon by means of ^{1}H and ^{13}C NMR. These dynamical heterogeneities make the precise determination of diffusion coefficients more difficult. This could explain the difference between close calculated diffusion coefficients of bulk and confined BMim Li TFSI at 300K and experimental shapes which show a slowdown of confined BMim cations.

Nevertheless, the characteristic residence time has shown an increasing trend upon confinement which is consistent with the decrease of Γ. Such behaviour is often observed in the nanoconfinement of molecular liquids [23,36] or ILs [15,16,40,41], and was nicely shown by NMR [41]. It is usually explained by the co-existence of two distinct populations of molecules under confinement: the first one, close to the confining matrix, corresponds to immobilized molecules interacting at the interface while the second one, far from the interface, has a more bulk-like behaviour [15,16,40,41]. This phenomenon could also occur for BMim Li TFSI confined in the biopolymer/silica network. However, this system involves two kinds of interpenetrated host networks confining three ionic species ($BMim^+$, Li^+ and $TFSI^-$ ions). $BMim^+$ cations could interact either with the biopolymer, or with the silica nanofibers network or with both of them. Guyomard-Lack et al have evidenced by means of Fourier transform infrared spectroscopy that an imidazolium-based IL has a preferential interaction with a tridimensional chitosan network [42]. The chemical structure of chitosan being very close to those of cellulosic derivatives like Si-HPMC, $BMim^+$ cations could favour interactions with the Si-HPMC matrix.

It is also worth pointing out that here, at each temperature, $\langle l \rangle$ increases together with τ_0 when the IL is confined inside the host network (Table 2). This indicates that confined $BMim^+$ cations diffuse less frequently but farther inside the ionogel. It is here interesting to consider in Table 2 the fitted parameters τ_0 and $\langle l \rangle$ around 300 K (290, 298, 300 K). τ_0 and $\langle l \rangle$ of confined BMim Li TFSI remained close to these of neat BMim TFSI (without lithium). It suggests that $BMIm^+$ cations in confined BMIm Li TFSI and in neat BMim TFSI have an analogous dynamics. As a consequence, regions far from interfaces could be more composed of BMim and TFSI ions and there can be competition between ionic species at the silica or biopolymer interfaces along with previous observation made above with chitosan matrix. This would be consistent with a segregation of lithium ions at silica interface. This preferential interaction of lithium with silica has already been demonstrated in several studies [37,43]. Our previous work[37] has already shown similar phenomenon in silica ionogels confining pyrrolidinium based IL with Li TFSI salt. In particular, the chemical shift of lithium observed by ^{7}Li MAS NMR increases from 0.9 ppm in bulk solution to 1.7 ppm when confined in silica ionogel. Besides, the Raman spectrum of bulk solution presents a shoulder at 749 cm^{-1} characteristic of the coordination of lithium with $TFSI^-$ anion, which decreases significantly in the case of silica ionogel. This indicates that lithium have less interactions with TFSI when confined inside silica ionogel. Moreover, Nordström et al [43] have studied dispersions of fumed silica in presence of BMim BF_4 with lithium salt. The authors also observed the preferential interaction of lithium with silica via electrostatic interactions with siloxane (Si-O-Si) groups. In the present work, the results obtained by QENS seem to indicate by an indirect way that lithium ions segregate at the interface of silica nanofibers, probably via electrostatic interactions with siloxane groups. This diffusion path for lithium ions together with the presence of a percolating silica network is consistent with the improved conductivity previously reported [12].

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Conclusions

Dynamical processes of BMim Li TFSI confined within interpenetrated biopolymer-silica networks was investigated by means of quasielastic neutron scattering, and compared to that of bulk counterpart. Similar information is also presented showing the effect of lithium salt on neat BMIm TFSI. In the space time scale investigated in this experiment (10 Å, 10 ps), one translational diffusion process was identified in bulk and confined cases.

For non confined IL-Li, $BMIm^+$ cation presents diffusion coefficients and characteristic times comparable to other ILs already reported in literature [15,28]. The presence of lithium ions has a considerable influence on the physical properties: disappearance of solid/liquid transition, decrease of diffusion coefficient of $BMim^+$, shorter characteristic time than in neat IL. Further studies with complementary energy ranges are however required to improve and deepen the understanding of their molecular motions.

Moreover, the confinement of IL with lithium salt within a biopolymer-silica matrix host network allowed reaching a very high loading of IL with a single-step procedure. At the molecular scale, dynamics of the $BMIm^+$ cation only slightly slowed down in the biopolymer-silica matrix compared to the bulk solution. This slowdown could be an average dynamics mainly due to the coexistence of two populations of cations in the material: the cations interacting with the confining network and the cations behaving like bulk IL far from interfaces. The segregation of lithium ions at the interface with the silica-based host network should also play a significant part on dynamics of $BMim^+$ cations inside the ionogel.

This ionogel based on biopolymer-silica interpenetrated networks retains liquid-like properties, a key point for applications in energy storage.

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